BARCS: beta-binomial regression for multivariable CRISPR screen designs

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Abstract

Background. Beta-binomial models represent a guide count as a proportion conditional on its library total. Existing CRISPR implementations of this idea are largely restricted to two-group comparisons, whereas modern screens often require time, dose, donor, or interaction effects.
Methods. BARCS implements guide-level beta-binomial regression, $\mathrm{logit}(\mu_{gi}) = \mathbf{x}_i^\top \boldsymbol{\beta}_g$. It conditions each guide count on the unfiltered library total, estimates prespecified coefficients or contrasts, and optionally uses negative-control calibration and dispersion moderation. Simulations with known truth and held-out controls were used to distinguish ranking performance from error-rate calibration.
Results. In a correctly specified continuous-dose simulation, unmoderated BARCS was anti-conservative: its null type-I error was 0.081 and its empirical false-discovery proportion was 0.130 at nominal 0.05, versus 0.038 and 0.049 for MAGeCK-MLE. A five-fold held-out analysis of 593 non-targeting guides in an IL2RA screen gave a BARCS $p < 0.05$ rate of 0.049 after control scaling, but the correlated FACS-bin design remains outside the independent-library model. A Cas13 sensitivity analysis used days 0, 7, and 14 in four replicate-complete cell lines; BARCS had a lower mean cell-line-specific null calibration error than MAGeCK-MLE (0.025 versus 0.057), whereas MAGeCK-MLE and general RNA-sequencing methods ranked essential genes more strongly. Because these inputs were normalized, ComBat-corrected, and rounded, that comparison does not validate either count likelihood. Dispersion and denominator ablations improved BARCS in simulations, but they do not establish superiority over competing methods.
Conclusions. BARCS supplies a design-matrix extension of beta-binomial CRISPR-screen analysis, but the present plug-in dispersion estimator does not provide uniformly calibrated finite-sample inference. Independent biological replication, likelihood-compatible counts, and held-out or cross-fitted controls are required. The results support the software contribution and define the work needed for confirmatory use; they do not show that beta-binomial regression dominates negative-binomial methods.

Keywords: CRISPR screen; beta-binomial regression; longitudinal design; count data; calibration

1 Introduction

Pooled CRISPR screens increasingly measure guide abundance across more than two experimental conditions. Examples include time courses, dose series, ordered FACS fractions, donor-stratified experiments, and factorial designs. In these studies the scientific estimand is a trend, an adjusted association, or an interaction. Reducing the experiment to a high-versus-low comparison discards intermediate observations and prevents direct adjustment for additional design variables.

The sampling model is consequential in this setting. RNA-sequencing methods typically model a transcript count with a negative-binomial distribution and represent library size through normalization or an offset [1–4]. Several

CRISPR-screen methods adopted this framework [5, 6]. A pooled-guide count also has a known denominator, however: the total number of mapped guide reads in that library. Conditional on this total, guide abundance is a proportion. The beta-binomial model represents this structure directly and separates depth-dependent sampling variation from heterogeneity among biological libraries. In the original CB^2 study, goodness-of-fit comparisons favored the beta-binomial over negative-binomial alternatives for pooled CRISPR counts [7].

Beta-binomial regression itself is not new. The `corncob` framework models both the mean and overdispersion of counts conditional on library totals [8], and general implementations are available in `aod`, `VGAM`, and `gamlss` [9–11]. BARCS is therefore not presented as the first beta-binomial regression method. Its contribution is a CRISPR-screen implementation that preserves immutable guide-library totals, uses an R design matrix, returns guide-level coefficient tests at screen scale, and connects this workflow to CB^2.

CB^2 applies its sampling model to two independent groups. For guide g observed in library i, let K_{gi} denote the guide count, N_i the unfiltered mapped-guide total, and $\mathbf{x}_i^\top$ one row of a prespecified, full-rank sample design matrix. The regression problem is

$$K_{gi} \mid N_i \sim \text{BetaBinomial}(N_i, \mu_{gi}, \rho_g), \qquad \text{logit}(\mu_{gi}) = \mathbf{x}_i^\top \boldsymbol{\beta}_g, \qquad H_0 : \mathbf{c}^\top \boldsymbol{\beta}_g = 0. \tag{1}$$

Here time, dose, donor, batch, or treatment is a sample-level predictor; the guide proportion remains the response. Independent biological libraries, not individual reads, determine the residual degrees of freedom.

Existing methods address parts of this problem. Repeated two-group analyses cannot estimate one coefficient conditional on the remaining columns of the design matrix. General RNA-sequencing tools support arbitrary designs but do not condition each guide on its observed library total. At the other end of the spectrum, Chronos and Waterbear model population dynamics or joint FACS-bin membership, respectively, but are tied to those experimental structures [12, 13]. A general-purpose, depth-aware coefficient test for pooled CRISPR designs remains missing.

We introduce BARCS (Beta-binomial Analysis and Regression for CRISPR Screens) to fill this gap. BARCS retains the beta-binomial sampling principle of CB^2, replaces the two group means with an R model matrix, and tests a prespecified coefficient or contrast with sample-level inference. The relationship to CB^2 is structural: both methods condition guide counts on library totals and cap the information contributed by deeper sequencing. Their estimators, effect scales, dispersion assumptions, and finite-sample reference distributions differ, so BARCS is not claimed to nest the implemented CB^2 test.

The empirical evaluation begins with calibration under known truth. It then uses the three-time-point Cas13 study of four replicate-complete cell lines from Liang et al. [14] as a processed-count sensitivity analysis and an ordered-bin IL2RA screen as a held-out-control diagnostic. Genome-scale and factorial simulations examine dispersion moderation and denominator choice. The serially harvested HT-29 culture, endpoint screens, and copy-number analyses are retained as descriptive robustness checks rather than evidence for sample-level longitudinal inference.

This design supports a deliberately limited claim. BARCS makes a library-total-conditional beta-binomial mean model available to designs that CB^2 cannot represent. Whether its coefficient probabilities are calibrated is a separate empirical question. The benchmarks therefore report failures as well as successes and do not treat the sampling-model choice as proof of superiority over negative-binomial regression.

2 Results

The Results begin with the nominal-versus-realized error-rate analysis because calibration is a prerequisite for interpreting any discovery comparison. We then state the limited relationship to CB^2, report the processed-count longitudinal and ordered-bin sensitivity analyses, and use controlled simulations to isolate dispersion and denominator choices. Endpoint, serial-harvest, copy-number, and external false-discovery audits are reported in the Supplement.

2.1 Finite-sample calibration is not guaranteed by the sampling model

The first analysis uses known truth rather than biological proxy labels. We simulated 200 genes with three guides per gene under the fitted beta-binomial model, including 160 null genes, 27 independent libraries, a continuous dose, and three batches. The tested model therefore matches the data-generating mean and variance families. Gene probabilities use the prespecified directional Stouffer summary for BARCS and the native gene statistic for MAGeCK-MLE.

Table 1: Nominal-versus-realized error rates in the continuous-dose beta-binomial simulation. Type-I error is the fraction of 160 null genes with $p < 0.05$. Empirical FDP is measured among calls at Benjamini–Hochberg FDR 0.05. This single deterministic simulation is a diagnostic, not a claim of uniform operating characteristics.

Method	Null type I	Power	Calls	Empirical FDP
Naive binomial z	0.844	1.000	174	0.770
BARCS unmoderated t	0.081	1.000	46	0.130
BARCS moderated t	0.088	1.000	44	0.091
Official MAGeCK-MLE Wald	0.038	0.975	41	0.049

BARCS is anti-conservative in this example despite correct beta-binomial simulation. The guide-level null rate is 0.077, so the failure is already present before gene aggregation. No fitted guide reaches the lower $\widehat{\rho} = 0$ boundary; the truncation mechanism is therefore not the cause in this simulation. More generally, the model-based covariance treats $\widehat{\rho}$ as fixed, and the residual-t reference does not propagate guide-specific dispersion uncertainty. Moderation lowers the empirical FDP but does not restore nominal type-I error.

Negative controls can diagnose this behavior, but evaluation must be separated from estimation. In the IL2RA FACS screen, 0.133 of 593 non-targeting guides have raw $p < 0.05$. The tail-quantile calibration sets its scale from the control tail, so measuring those same controls after calibration would be circular. Five-fold cross-fitting instead estimates each scale on four folds and scores the omitted fold. The resulting held-out rate is 0.049. This is a valid out-of-sample diagnostic of the scaling rule at the selected threshold; it does not repair the correlation among bins from one donor or validate the independent-library likelihood.

The processed Liang analysis provides a secondary, weaker diagnostic. BARCS has a macro-averaged null $p < 0.05$ rate of 0.064. Its reported calibration error, 0.025, is the mean of the four cell-line-specific absolute errors, not $|0.064 - 0.05| = 0.014$. MAGeCK-MLE has corresponding values 0.107 and 0.057. Because the common matrix was normalized, ComBat-corrected, and rounded before either fit, these values compare procedures on that transformed input but cannot establish calibration of their count likelihoods.

2.2 BARCS generalizes the CB^2 sampling model

CB^2 analyzed a guide count as a proportion conditional on the corresponding library total. This conditioning matters because the beta-binomial variance separates two sources of uncertainty: binomial sampling variation, which decreases with sequencing depth, and between-library heterogeneity, which does not. A negative-binomial model can accommodate overdispersion in marginal counts, but it does not encode the observed library total as the denominator of the guide proportion. The original article reported goodness-of-fit advantages over negative-binomial alternatives [7]; the present benchmarks do not repeat that distributional comparison. They ask whether the library-total-conditional model can be implemented for regression designs beyond the two groups handled by CB^2.

BARCS changes the mean model while retaining the sampling model. For guide g in library i,

$$K_{gi} \mid N_i \sim \mathrm{BetaBinomial}(N_i, \mu_{gi}, \rho_g), \qquad \mathrm{logit}(\mu_{gi}) = \mathbf{x}_i^\top \boldsymbol{\beta}_g.$$

The original two-group question is recovered when $\mathbf{x}_i$ contains an intercept and one group indicator. More generally, the columns of $\mathbf{X}$ can represent time, dose, donor, batch, an interaction, or a prespecified contrast. Thus BARCS generalizes the form of the mean model; it is not claimed to reproduce or outperform CB^2 on the binary design for which the original method was developed. The two implementations use different effect scales, dispersion estimates, working weights, and reference degrees of freedom.

2.3 Processed Cas13 data provide a longitudinal sensitivity analysis

The Cas13 fitness screens of Liang et al. [14] contain measurements from days 0, 7, and 14. K562 was excluded because one day-0 replicate was not deposited. In the four retained cell lines, BARCS, MAGeCK-MLE, edgeR-QL, DESeq2, and limma–voom estimated the same continuous time slope with a replicate block and used two-sided significance tests.

The deposited values had already undergone median-of-ratios normalization, ComBat correction, and replicate-outlier processing. We rounded them once to the nearest pseudo-count and supplied the same matrix to every method. Accordingly, this is a processed-count sensitivity analysis rather than a raw-count likelihood comparison. In particular, it cannot be used as primary evidence that a beta-binomial likelihood is better calibrated than a negative-binomial likelihood.

Across cell lines, the mean null $p < 0.05$ rate was 0.064 for BARCS and 0.107 for MAGeCK-MLE. The corresponding mean cell-line-specific absolute calibration errors were 0.025 and 0.057; 0.025 is not the absolute deviation of the macro-averaged 0.064 rate. MAGeCK-MLE ranked essential genes more strongly (average precision 0.877 versus 0.838), as did edgeR-QL, DESeq2, and limma–voom (0.874–0.876).

At an empirical 5% null false-positive rate, essential-gene recall was 0.863 for BARCS and 0.879–0.888 for the alternative longitudinal regression methods. Figure 1 shows the complete method-specific volcano plots across all four retained cell lines and three guide trajectories from HAP1, THP1, and MDA-MB-231. These guides target lncRNAs assigned to the proxy-null set because their measured expression was zero. BARCS returned $p = 0.20$–0.23, whereas DESeq2, edgeR-QL, and limma–voom returned $p < 0.05$. The examples illustrate disagreement on the transformed matrix; they are not independent validation of the guides' biological null status.

Rounding changed each deposited value by less than 0.5, but the upstream transformations mean that none of the fitted likelihoods retains its literal raw-count interpretation. Confirmation from raw reads therefore remains necessary before drawing a distributional or confirmatory inference claim. The supplied streaming workflow performs the full longitudinal preparation for that validation.

2.4 A covariate-adjusted ordered phenotype recovers validated regulators

We next asked whether the design-matrix extension is useful outside a time course. GSE242880 contains an IL2RA protein-abundance screen in primary human T cells, with four ordered FACS bins for each of three donors, 6,000 guides, and 26 regulators validated by individual knockout and flow cytometry [13, 15]. The four bins were assigned prespecified latent normal locations, and BARCS fitted one marker-abundance slope with donor indicators. Official MAGeCK-MLE received the same design.

This experiment also provides a direct model diagnostic. Bins from one donor partition the same cell pool and are negatively dependent, whereas the guide-level BARCS fit treats their margins independently. Among 593 non-targeting guides, the uncalibrated $p < 0.05$ frequency was 0.133. Because `bb_calibrate_controls()` estimates its scale from the control tail, evaluating those same guides after fitting would be circular. We therefore used deterministic five-fold cross-fitting: each guide was evaluated with a scale estimated from the other four folds. The held-out $p < 0.05$ rate was 0.049. The production gene analysis used all controls to estimate one scale; calibration does not change effect estimates or their ranking.

Control-calibrated BARCS recovered 22 of 26 validated regulators with 49 discoveries, compared with 17 of 26 and 72 discoveries for the matched MAGeCK-MLE fit (Table 2). The outer-bin ablation recovered 18 regulators under both BARCS and `mageck test`. The intermediate bins therefore contributed information that was lost by the binary comparison. In the selected 33-gene experimental follow-up panel, calibrated BARCS had F1 0.863, Matthews correlation 0.398, and balanced accuracy 0.709; the corresponding MAGeCK-MLE values were 0.723,

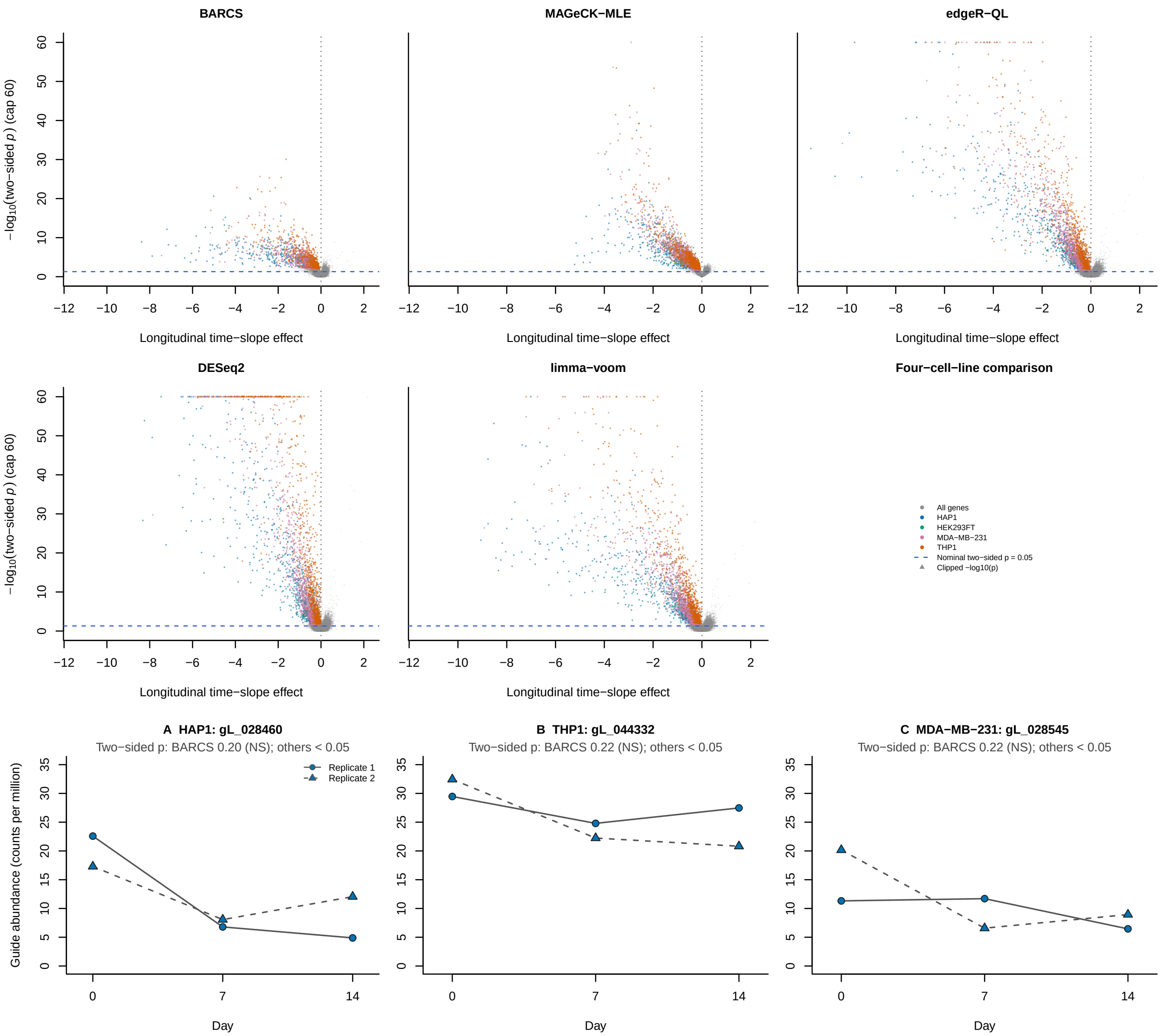

Figure 1: Longitudinal Cas13 sensitivity analysis using days 0, 7, and 14. The upper rows show method-specific volcano plots for BARCS, MAGeCK-MLE, edgeR-QL, DESeq2, and limma–voom, pooling gene-level results from HAP1, HEK293FT, MDA-MB-231, and THP1. K562 is excluded because one day-0 replicate is absent. Within each retained cell line, every method estimates the same continuous time slope from the identical processed-count matrix. Colored points are negatively directed genes called at two-sided FDR 0.10, with color denoting cell line; the dashed horizontal line marks nominal two-sided $p = 0.05$, and triangles identify $-\log_{10}(p)$ values clipped at 60. (A–C) Processed guide counts divided by the corresponding library size, in counts per million, for three proxy-null examples. Lines connect measurements from the same replicate across days. BARCS leaves each guide non-significant, whereas DESeq2, edgeR-QL, and limma–voom each call it significant under the proxy-null benchmark. MAGeCK-MLE is included in the gene-level volcano comparison but not in panels A–C because its reported statistic is gene-level.

Table 2: Ordered-bin IL2RA sensitivity analysis. Recovery requires a discovery at each method's threshold and the experimentally validated direction. The BARCS NTC rate is five-fold held out; it is not measured on the controls used to estimate that fold's scale.

Method	Discoveries	Validated recovery	NTC $p < 0.05$
BARCS, four bins + controls	49	22/26	0.049
BARCS, four bins raw	127	23/26	0.133
BARCS, outer bins	34	18/26	0.064
MAGeCK-MLE, four bins	72	17/26	—
MAGeCK test, outer bins	30	18/26	—
Waterbear, four bins [13]	79	24/26	—
MAUDE, four bins [16]	406	25/26	—

0.070, and 0.541. Because the panel was selected for follow-up from a previous screen, these metrics are supporting evidence rather than unbiased genome-wide accuracy estimates.

The specialist models remain important comparators. Waterbear recovered 24 validated regulators by modeling joint bin probabilities, guide-to-gene structure, and uncertain cutoffs. MAUDE recovered 25 but called 406 of 1,350 genes; in a separately held-out control analysis its empirical false-positive rate was 0.095 at a nominal 0.05. BARCS is therefore not the most faithful generative model for sorted bins. Its contribution is a transparent, general-purpose coefficient analysis that retains the ordered phenotype and donor adjustment. Negative controls expose the independence violation and can scale one tail operating point, but they do not make the correlated-bin model correct.

2.5 Simulation identifies an internal benefit of dispersion moderation

We used CRISPulator to evaluate a FACS design under known truth [17]. Three prespecified genome-scale seeds each generated 10,000 genes, five guides per gene, four independent replicates, and low, bulk, and high fractions. Twenty percent of genes affected the phenotype and 5% were explicit negative controls. All fitted methods received the same simulated counts and design.

The genome-scale analysis was designed as an ablation within BARCS. BARCS-ST uses the original guide-specific dispersion estimate, whereas BARCS-MOD applies the scaled-F moderation defined in Methods before using the same gene summary. Moderation increased average precision from 0.902 to 0.919 and F1 at gene FDR 0.10 from 0.811 to 0.847; realized FDP was 0.065 and 0.066, respectively. Because the effects, gene aggregation, controls, and thresholds are shared, this supports a benefit of moderation within BARCS. BARCS-MOD was not applied to the real-data benchmarks, so this result does not explain or repair their calibration behavior.

MAGeCK-MLE and CRISPhieRmix were included in the 10,000-gene run, but that restricted comparator set is not sufficient for a general method-ranking claim. A separate five-seed, 400-gene baseline included edgeR-QL, DESeq2, and limma–voom. Their mean AUROCs were 0.965, 0.963, and 0.965, compared with 0.955 for BARCS-ST and 0.964 for MAGeCK-MLE. Mean F1 ranged only from 0.817 to 0.832 across these five count models, while realized FDP ranged from 0.048 for MAGeCK-MLE and 0.086 for BARCS-ST to 0.218–0.240 for the three general RNA-sequencing methods. Thus ranking, F1, and calibration favor different methods, and no count model dominates this broader comparison.

Supplementary Figure S2 shows F1 and realized FDP across requested thresholds for the restricted genome-scale set. Curves matched by realized FDP are useful for the BARCS-ST versus BARCS-MOD ablation, but they must not be read as evidence that BARCS-MOD leads methods omitted from that run.

2.6 Interaction recovery: what the denominator decides

The FACS simulation gives every guide one effect. The design BARCS was built for has more than one, so we also simulated a factorial screen with simCRISPR [18]: knockout induction crossed with treatment, propagated over five days, with amplification and sequencing applied afterwards. Each sgRNA then carries a knockout effect, a treatment effect, and an interaction between them, and the interaction is a coefficient in a design matrix rather than a difference of two arms. BARCS tests it directly, as β_3 in $\text{logit}(\mu_{gi}) = \beta_0 + \beta_1 k_i + \beta_2 t_i + \beta_3 k_i t_i$.

Two features of the simulator make the scoring unusually literal. Non-targeting guides have an interaction of exactly zero, so the null is ground truth and not an empirical stand-in; and safe-harbor guides also have zero targeted interaction while still carrying the DNA-damage response of being cut, which separates *no effect* from *no gene effect*. Each control class was split in half so that the guides used to normalize and calibrate are never the guides scored.

Table 3: Interaction recovery on simCRISPR screens: 2,000 sgRNAs, three replicates per factorial arm, and three seeds (mean). Recall and F1 use targeting guides with $|\text{interaction}| \geq 0.2$ as positives. Null rate is the fraction of held-out non-targeting guides called at FDR 0.10; safe-harbor rate is the corresponding rate among cut-but-inactive guides.

Method	Denominator	Dir. recall	F1	Null rate	Safe-harbor
■ BARCS-ST	Library	0.376	0.488	0.004	0.002
■ BARCS-MOD	Library	0.375	0.481	0.000	0.000
■ BARCS-ST	Non-targeting	0.769	0.870	0.024	0.070
■ BARCS-MOD	Non-targeting	**0.838**	**0.917**	0.040	0.080
■ BARCS-ST	Safe-harbor	0.762	0.864	0.042	0.042
■ BARCS-MOD	Safe-harbor	0.830	0.914	0.027	**0.035**
■ MAGeCK-MLE	Non-targeting	0.003	0.006	0.000	0.000

Effect recovery is insensitive to the denominator; detection is not. Effect recovery is constant at $\rho_s = 0.687$–0.690 across all three denominators, because a shared multiplicative shift displaces all guides together and leaves their relative order almost unchanged. The detection metrics differ substantially. Under the full-library denominator BARCS-MOD attains directional recall 0.375 and F1 0.481; under either control denominator it attains 0.830–0.838 and 0.914–0.917. The difference therefore originates in the variance model rather than in the point estimates.

The mechanism is evident in the final column. Thirty-eight percent of the targeting guides are at least halved in the treated knockout arm, which inflates the share of every surviving guide and thereby confers an apparent interaction on each: across the true zeros the median estimated coefficient is +0.18 where the true value is 0, and its interquartile range excludes zero. The control calibration absorbs this excess as designed, inflating the empirical null approximately twofold, to 1.99 and 2.24. The library rows are consequently not anti-conservative—their null rates of 0.004 and 0.000 fall well below the nominal 0.10—but they are correspondingly underpowered, the bias having been transferred into lost power rather than into false positives. The control denominators require essentially no inflation, 1.00 to 1.07, and retain power.

Safe-harbor and non-targeting controls are not interchangeable. The two control denominators are indistinguishable on detection: AUROC 0.953 against 0.957, F1 0.914 against 0.917. They differ on which errors survive. Normalizing on non-targeting guides leaves the safe-harbor guides called at 0.080, against 0.035 when the safe-harbor guides set the denominator themselves. Safe-harbor guides are cut and non-targeting guides are not, so only the former carry the DNA-damage response that the targeting guides also carry; a denominator built from guides that were never cut cannot remove it. This reproduces, using an independent inference procedure, the conclusion that simCRISPR was developed to establish, and it permits a more specific recommendation: the non-targeting denominator is preferable for detection, the safe-harbor denominator for suppressing cutting artifacts,

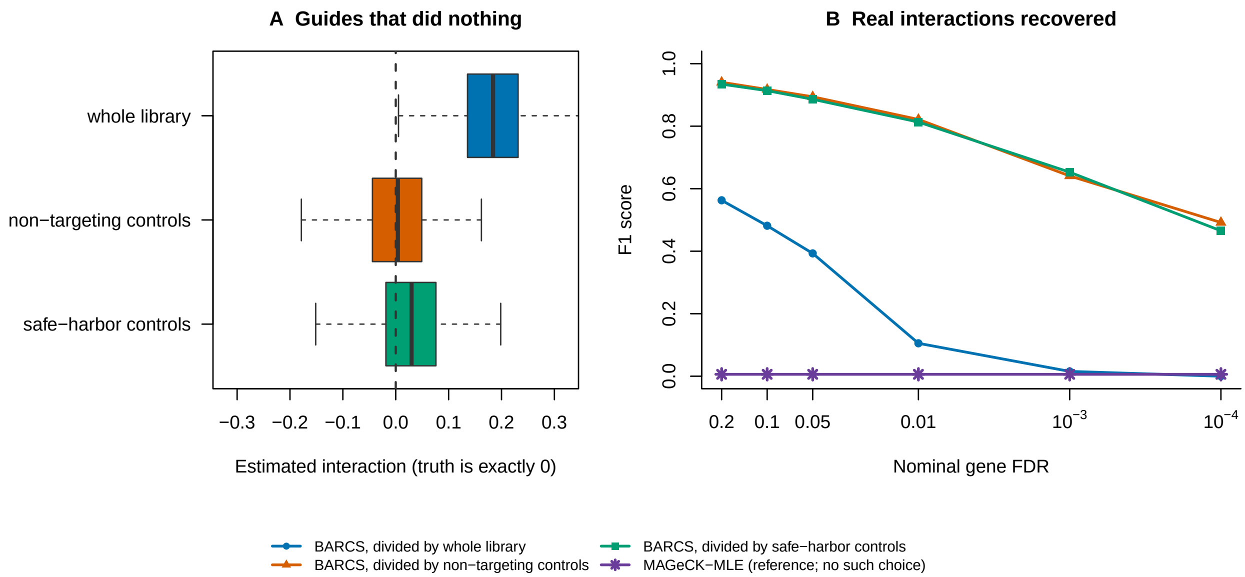

Figure 2: Denominator choice controls interaction recovery. (A) Estimated interactions among true-zero guides. A full-library denominator induces a positive shift because widespread depletion changes library composition; either control denominator removes it. (B) F1 across requested FDR thresholds. Non-targeting and safe-harbor denominators retain substantially more power than the full-library denominator. MAGeCK-MLE is shown only as a reference because the simulated estimand is guide-level.

and the difference between their safe-harbor rates quantifies the penalty for selecting the wrong one.

Moderation confers the same benefit here as at genome scale, and for the same reason: F1 rises from 0.870 to 0.917 under the non-targeting denominator and from 0.864 to 0.914 under safe-harbor, with the null rate still under the nominal target. Across cutoffs the ordering is stable, with BARCS-MOD F1 running 0.492, 0.641, 0.822, 0.894, and 0.917 from a requested 10^{-4} to 0.10 while the held-out null rate stays at or below the cutoff throughout.

This design lies outside the intended scope of MAGeCK-MLE. Its ranking performance is moderate, at AUROC 0.854, and its effect correlation is the highest observed, at 0.698, but it returns 1.7 calls from approximately 500 substantial interactions and its F1 is 0.006 at every cutoff from 10^{-4} to 0.20. This reflects a mismatch of estimand rather than a defect exposed by the simulation: the simulated truth is guide-level, so each sgRNA constitutes its own gene, which leaves a gene-level model no guides across which to pool and restricts its permutation null to 2,000 draws. We report the result to delimit the applicable range rather than to rank the method, and it is the quantization limit documented at genome scale, encountered from the opposite direction. An estimand defined per guide requires a guide-level test.

3 Methods

3.1 From CB^2 to BARCS

BARCS is fundamentally a guide-level coefficient model. Its required inputs are a guide-by-library count matrix, the corresponding full-library totals, and a sample design matrix; its primary output is one coefficient or contrast test per guide. Gene aggregation is downstream and optional. In particular, neither Fisher nor Stouffer combination is required to fit or interpret the BARCS regression. Some benchmarks below use an explicitly labeled Stouffer summary to compare with deposited gene-level results, whereas BARCS-GC uses the direct shared-effect Wald construction in Equations (30)–(33).

CB^2 adapts the beta-binomial weighted test of Baggerly et al. [19] to CRISPR pooled screens [7]. Its central insight is worth preserving: sequencing depth measures a guide proportion precisely, but it does not turn reads into independent biological replicates. The original method applies this insight to a two-group contrast. BARCS retains the same depth-aware, extra-binomial variance principle and changes the mean model from two group means to a design matrix. “Retains” refers to the stochastic hierarchy and the sample-level information ceiling; it does not mean that the default logit fit reproduces the original finite-sample statistic. The differences and the limited local relationship are stated explicitly below.

Table 4: What changes—and what does not—from the original CB^2 method to BARCS.

Feature	Original CB^2	BARCS
Primary question	Difference between two groups	Trend, adjusted association, interaction, or contrast
Sample design	Reference and comparison labels	Arbitrary full-rank R model matrix
Mean model	Two unrelated group proportions	$\mathrm{logit}(\mu_i) = \mathbf{x}_i^\top \boldsymbol{\beta}$
Effect	Difference or fold change between groups	Conditional log-odds coefficient or linear contrast
Dispersion	Separate beta-binomial weighting within each group	One guide-specific ρ shared across the fitted design
Estimator	Weighted group proportions	Feasible logistic IRLS
Variance principle	Binomial sequencing plus between-library beta-binomial heterogeneity	
Reference df	Group-variance df formula	Residual $m - q$
Adjustment	Pairwise stratification or preprocessing	Batch, donor, dose, time, interactions, splines, and other covariates
Implementation	Existing CB^2 workflow	Regression API with optional RcppArmadillo kernels

The regression bridge was described by Baggerly et al. [20] for sequencing count data with multiple groups and continuous covariates. The method combines logistic regression with extra-binomial variation and retains a coefficient-based t test. Here we state that construction in CRISPR-screen notation and use the beta-binomial, library-size-dependent variance of Williams [21]. An identity-link regression of raw proportions would allow fitted values outside $(0, 1)$, while an unweighted correlation would ignore unequal library sizes. The logit and beta-binomial weights solve those two problems directly.

3.2 Model

For guide g in sequenced sample i, let K_{gi} be its read count, let N_i be the full mapped-guide total for that sample, and let $Y_{gi} = K_{gi}/N_i$ be the observed guide proportion. Sample-level quantities such as dose, time, treatment, donor, and batch are predictors even when they are described biologically as experimental “outcomes.”

The model is fitted separately for every guide, so the guide index is suppressed below. Conditional on a latent sample-specific abundance P_i,

$$K_i \mid P_i \sim \mathrm{Binomial}(N_i, P_i), \tag{2}$$

$$P_i \sim \mathrm{Beta}(\mu_i \kappa, (1 - \mu_i)\kappa). \tag{3}$$

The mean is linked to a row vector of sample covariates $\mathbf{x}_i^\top$ by

$$\mathrm{logit}(\mu_i) = \log\left(\frac{\mu_i}{1 - \mu_i}\right) = \mathbf{x}_i^\top \boldsymbol{\beta}. \tag{4}$$

The design can contain an intercept, a continuous phenotype, batch indicators, interactions, spline terms, and other prespecified adjustment variables.

It is useful to express the beta precision κ as an intraclass correlation

$$\rho = \frac{1}{\kappa + 1}, \qquad 0 \le \rho < 1. \tag{5}$$

Marginally,

$$\begin{aligned} \mathrm{E}(K_i) &= N_i \mu_i, &(6)\\ \mathrm{Var}(K_i) &= N_i \mu_i (1 - \mu_i)\{1 + (N_i - 1)\rho\}, &(7)\\ \mathrm{Var}(Y_i) &= \mu_i (1 - \mu_i) \left\{ \frac{1-\rho}{N_i} + \rho \right\}. &(8) \end{aligned}$$

Equation (8) cleanly separates sequencing uncertainty, which decreases with N_i, from between-sample heterogeneity, which does not vanish when sequencing depth becomes arbitrarily large.

3.2.1 Interpretation of the continuous coefficient

Suppose the model contains a continuous phenotype d_i with coefficient β_d. Then $\exp(\beta_d)$ is the multiplicative change in the odds of observing that guide for a one-unit increase in d_i, conditional on the other covariates. Standardizing d_i before fitting makes $\exp(\beta_d)$ the odds ratio per sample standard deviation and makes guide effects directly comparable. For rare guides, odds and proportions are similar, so β_d is also approximately a log abundance ratio per unit of d_i.

3.3 Estimation by feasible IRLS

Let X be the $m \times q$ full-rank design matrix. Given current values of $\boldsymbol{\beta}$ and ρ, define

$$\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}, \qquad \mu_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}, \tag{9}$$

$$z_i = \eta_i + \frac{Y_i - \mu_i}{\mu_i(1 - \mu_i)}, \qquad w_i = \frac{N_i \mu_i (1 - \mu_i)}{1 + (N_i - 1)\rho}. \tag{10}$$

The coefficient update is the weighted least-squares step

$$\widehat{\boldsymbol{\beta}}_{\text{new}} = (X^\top W X)^{-1} X^\top W \mathbf{z}, \qquad W = \mathrm{diag}(w_1, \ldots, w_m). \tag{11}$$

This is weighted least squares on the logistic working response, not on the raw proportions.

For a fixed fitted mean, ρ is estimated from the Pearson equation

$$\sum_{i=1}^{m} \frac{(K_i - N_i \widehat{\mu}_i)^2}{N_i \widehat{\mu}_i (1 - \widehat{\mu}_i)\{1 + (N_i - 1)\widehat{\rho}\}} = m - q. \tag{12}$$

If the left side at $\rho = 0$ is no larger than $m - q$, the estimate is truncated at zero rather than claiming negative overdispersion. Equations (10)–(12) are alternated to convergence.

The weight has an important limiting behavior:

$$\lim_{N_i \to \infty} w_i = \frac{\mu_i (1 - \mu_i)}{\rho} \quad \text{when } \rho > 0. \tag{13}$$

Therefore a deeply sequenced sample cannot acquire unlimited leverage. Once sequencing uncertainty is small, biological heterogeneity sets the information ceiling.

3.4 T-based inference

At convergence, the model-based covariance estimator is

$$\widehat{\mathrm{Cov}}(\widehat{\boldsymbol{\beta}}) = (X^\top \widehat{W} X)^{-1}. \tag{14}$$

The Pearson equation scales the residual variation to approximately one. If the dispersion estimate reaches its numerical upper boundary, the implementation additionally multiplies Equation (14) by the remaining Pearson scale.

Equation (14) is a plug-in, model-based covariance: it conditions on the estimated $\widehat{\rho}$ as though it were fixed. It does not propagate uncertainty from estimating a separate dispersion for every guide. Referring the coefficient to t_{m-q} gives a heavier-tailed small-sample reference than a normal approximation, but it is not a formal correction for dispersion-estimation uncertainty. With few independent libraries, calibration must therefore be checked by simulation, phenotype permutation, or prespecified negative controls.

For coefficient β_j, the statistic

$$t_j = \frac{\widehat{\beta}_j - \beta_{j,0}}{\sqrt{\widehat{\mathrm{Cov}}(\widehat{\boldsymbol{\beta}})_{jj}}} \tag{15}$$

is compared with a Student t_{m-q} distribution. The degrees of freedom are driven by the number of independent sequenced samples, not by the total read count. This is the crucial reason that a library with millions of reads does not create artificial certainty.

More generally, for a prespecified contrast vector $\mathbf{c}$,

$$t_{\mathbf{c}} = \frac{\mathbf{c}^\top \widehat{\boldsymbol{\beta}} - \delta_0}{\sqrt{\mathbf{c}^\top \widehat{\mathrm{Cov}}(\widehat{\boldsymbol{\beta}})\mathbf{c}}} \tag{16}$$

uses the same $m - q$ degrees of freedom. This covers pairwise group contrasts, average slopes in interaction models, and other one-degree-of-freedom hypotheses. Guide-wise two-sided p-values are adjusted using the Benjamini–Hochberg procedure [22].

3.4.1 Optional guide-dispersion moderation

BARCS-MOD is an optional empirical-Bayes transformation of the fitted guide statistics, not a separate likelihood. Let $\nu = m - q$ and let $s_g^2 = Q_g^{(0)}/\nu$ be guide g's Pearson variance inflation under the binomial working model. A lowess trend in log mean CPM gives a guide-specific prior scale s_{0g}^2, and the scaled-F log-moment estimator of Smyth [23] supplies prior degrees of freedom d_0. The moderated inflation is

$$\widetilde{s}_g^2 = \frac{\nu s_g^2 + d_0 s_{0g}^2}{\nu + d_0}. \tag{17}$$

Because the original beta-binomial fit cannot estimate negative overdispersion, its stored standard error corresponds to the fitted inflation $s_{g,\mathrm{fit}}^2 = \max(1, s_g^2)$. The implementation therefore reports

$$\mathrm{SE}_{g,\mathrm{mod}} = \mathrm{SE}_g \sqrt{\frac{\widetilde{s}_g^2}{\max(1, s_g^2)}}, \qquad t_{g,\mathrm{mod}} = \frac{\widehat{\beta}_g}{\mathrm{SE}_{g,\mathrm{mod}}}, \qquad \nu_{\mathrm{mod}} = \nu + d_0. \tag{18}$$

The one-sided `pmax` truncation in Equation (18) is part of the implementation and is not an estimated scientific constraint. The optional `one_way=TRUE` variant additionally replaces $\widetilde{s}_g^2$ by $\max(\widetilde{s}_g^2, s_{g,\mathrm{fit}}^2)$, so moderation can only increase a standard error. All reported BARCS-MOD results are simulation ablations; the real-data benchmarks use unmoderated BARCS.

3.4.2 From guide statistics to gene statistics

BARCS fits the count model guide by guide. The FACS, simulation, GSE70038, A375, and descriptive HT-29 benchmark scripts then use an explicit directional Stouffer summary, denoted BARCS-ST below, to obtain one gene-level statistic. For the jth valid guide targeting gene g, the two-sided guide p-value is converted back to a signed standard-normal score,

$$z_{gj} = \operatorname{sign}(\widehat{\beta}_{gj})\Phi^{-1}\left(1 - \frac{p_{gj}}{2}\right). \tag{19}$$

If m_g valid guides target the gene, their combined score and two-sided gene-level p-value are

$$Z_g = \frac{1}{\sqrt{m_g}} \sum_{j=1}^{m_g} z_{gj}, \qquad p_g = 2\Phi(-|Z_g|). \tag{20}$$

The reported gene effect is the median of the guide coefficient estimates,

$$\widehat{\beta}_g = \operatorname{median}_{j=1,\dots,m_g} \widehat{\beta}_{gj}, \tag{21}$$

and gene-level false-discovery rates are obtained by applying Benjamini–Hochberg to the collection of p_g values. Consequently, concordant guide directions reinforce one another, discordant directions cancel, and one extreme guide has limited influence on the reported effect.

This aggregation is deliberately transparent, but it is not a native hierarchical gene model. Equations (19)–(20) give equal weight to valid guides and use the independence reference $\sqrt{m_g}$; shared sequence artifacts or other correlation can therefore make the gene statistic anti-conservative. A production extension should estimate guide reliability and within-gene dependence or use a hierarchical effect model. In the biological head-to-head benchmarks, this Stouffer–median procedure is used only for BARCS (and for the deliberately naive binomial comparator). MAGeCK, Chronos, BAGEL2, Waterbear, and MAUDE retain their native or deposited gene-level summaries. The exploratory DeWeirdt false-discovery audit used Fisher aggregation as stated in its Results description and is not pooled with these primary benchmark statistics.

Exchangeable normal guide-beta statistic. BARCS-NORM implements a different and deliberately simple model. It does not transform guide p-values and does not use guide-specific standard errors. Instead, for the m_g valid guides targeting gene g, it assumes

$$\widehat{\beta}_{gj} \overset{\text{iid}}{\sim} N(\mu_g, \sigma_g^2). \tag{22}$$

The gene effect and between-guide standard deviation are estimated directly:

$$\widehat{\mu}_g = \frac{1}{m_g} \sum_{j=1}^{m_g} \widehat{\beta}_{gj}, \tag{23}$$

$$s_g^2 = \frac{1}{m_g - 1} \sum_{j=1}^{m_g} (\widehat{\beta}_{gj} - \widehat{\mu}_g)^2. \tag{24}$$

Testing $H_0 : \mu_g = 0$ gives

$$T_g = \frac{\widehat{\mu}_g}{s_g/\sqrt{m_g}} \sim t_{m_g-1} \quad \text{under Equation (22) and } H_0. \tag{25}$$

Thus normally distributed guide betas lead to a Student t reference when their unknown σ_g is estimated from the same guides. The software also reports $2\Phi(-|T_g|)$ as a plug-in normal sensitivity value, but it is not the primary p-value because it treats the estimated spread as known. Multiple guides support this guide-consistency calculation; they do not become additional biological screen replicates.

Random-effects summaries. BARCS-RE (the implementation label `BARCS-partial`) retains each guide-specific regression standard error and models

$$\widehat{\beta}_{gj} \mid \beta_g, \tau_g^2 \sim N\Big(\beta_g,\ \mathrm{SE}(\widehat{\beta}_{gj})^2 + \tau_g^2\Big). \tag{26}$$

It estimates τ_g^2 by the nonnegative DerSimonian–Laird estimator [24] and then pools guide effects with weights

$$w_{gj} = \Big\{\mathrm{SE}(\widehat{\beta}_{gj})^2 + \widehat{\tau}_g^2\Big\}^{-1}, \qquad \widehat{\beta}_g = \frac{\sum_j w_{gj}\widehat{\beta}_{gj}}{\sum_j w_{gj}}. \tag{27}$$

BARCS-RE-EB (implementation label `BARCS-EB`) stabilizes the noisy gene-specific heterogeneity estimate by shrinking it toward the pooled screen-wide excess-Q estimate τ_0^2:

$$\widetilde{\tau}_g^2 = \frac{(m_g-1)\widehat{\tau}_g^2 + d_0\tau_0^2}{(m_g-1)+d_0}, \qquad d_0 = 4. \tag{28}$$

Both random-effects statistics are calibrated against prespecified negative-control genes when at least ten are available and otherwise against a robust whole-screen empirical null. In contrast, BARCS-NORM obtains its spread and reference degrees of freedom entirely within each gene. These three alternatives and BARCS-ST reuse exactly the same fitted guide coefficients; only the guide-to-gene inference changes.

Guide-consistency statistic for the one-replicate boundary. When sample replication is insufficient, converting a guide statistic through a low-degree-of-freedom t distribution can erase useful ranking information. We therefore prespecified a separate exploratory gene statistic, BARCS-GC, for the $R = 1$ CRISPulator analysis. It uses the uncalibrated guide coefficient and model-based standard error to estimate one shared gene effect. With

$$w_{gj} = \mathrm{SE}(\widehat{\beta}_{gj})^{-2}, \tag{29}$$

the inverse-variance estimate, its standard error, and raw Wald statistic are

$$\widehat{\beta}_g = \frac{\sum_{j=1}^{m_g} w_{gj}\widehat{\beta}_{gj}}{\sum_{j=1}^{m_g} w_{gj}}, \qquad \mathrm{SE}(\widehat{\beta}_g) = \left(\sum_{j=1}^{m_g} w_{gj}\right)^{-1/2}, \qquad T_g = \frac{\widehat{\beta}_g}{\mathrm{SE}(\widehat{\beta}_g)}. \tag{30}$$

This is a direct fixed-effect estimate of the common guide coefficient, not a Fisher or Stouffer combination of guide p-values. Its additional assumption is that the guide coefficients target one common gene effect; the reported direction-agreement fraction exposes gross heterogeneity.

Because T_g does not have a trustworthy sample-replicate reference distribution at $R = 1$, it is calibrated across genes. Let c_0 be the all-gene median and

$$s_{\mathrm{MAD}} = \frac{\mathrm{median}_g\, |T_g - c_0|}{\Phi^{-1}(0.75)}. \tag{31}$$

When at least ten valid control genes are present, let c_C be their median and let

$$s_C = \frac{Q_{0.95}\{|T_g - c_C| : g \in C\}}{\Phi^{-1}(0.975)}. \tag{32}$$

The calibrated statistic and two-sided empirical-null p-value are

$$Z_g^{\mathrm{GC}} = \frac{T_g - c_C}{\max(1, s_{\mathrm{MAD}}, s_C)}, \qquad p_g^{\mathrm{GC}} = 2\Phi(-|Z_g^{\mathrm{GC}}|), \tag{33}$$

followed by Benjamini–Hochberg adjustment. If too few control genes are available, c_0 replaces c_C and s_C is omitted. At least three finite guide coefficient estimates are required per gene. The all-gene MAD assumes that fewer than half of tested genes are active.

BARCS-GC measures reproducibility across independently designed perturbations; it does not turn guides into independent biological samples. Its p-value is consequently labeled empirical-null guide-consistency evidence, not sample-level confirmation. The ordinary BARCS result remains the primary analysis when biological replicates are available, and independent experimental replication remains necessary for confirmatory claims.

3.4.3 Continuous-dose calibration simulation

The nominal-error diagnostic in Table 1 uses seed 20260723 and contains 200 genes, three independently simulated guides per gene, and 27 libraries formed by nine dose values crossed with three batches. Of the genes, 160 are null and 40 have dose coefficients +0.9 or −0.9. Library totals are sampled uniformly from 40,000 to 120,000 reads. Guide baselines are uniform on the logit scale from −7.4 to −6.5, the two nonreference batches have coefficients 0.20 and −0.15, and counts are drawn from the beta-binomial hierarchy with $\rho = 0.0015$.

BARCS fits `~dose + batch`; BARCS-MOD then applies Equations (17)–(18). The naive comparator is a binomial logistic regression, and official MAGeCK-MLE 0.5.9.5 receives the same design matrix. BARCS and the naive comparator use the same directional Stouffer gene summary, while MAGeCK-MLE retains its native gene result. Type-I error is the unadjusted $p < 0.05$ fraction among the 160 null genes. Empirical FDP is the null fraction among genes called at Benjamini–Hochberg FDR 0.05. The executable simulation, compact metrics, and dispersion-boundary diagnostics are included with the deposited analysis.

3.4.4 Longitudinal HT-29 benchmark

Guide counts for the HT-29 time course of Tzelepis et al. [25] were obtained from Data S2 of that article through the Europe PMC supplementary bundle for PMC5081405. The three sequencing columns reported at each of days 7, 10, 13, 16, 19, 22, and 25 were summed to one column per harvest and combined with the pDNA measurement, and guides with fewer than 30 pDNA reads were removed, leaving 86,882 guides targeting 17,995 genes. Library totals were computed before that filter and never recomputed, so the beta-binomial denominator remains the full mapped-read total for each sample; the filter removes 4.2% of guides but only 0.10% of reads, uniformly across samples. BARCS and official MAGeCK-MLE 0.5.9.5 were both fitted to this eight-column matrix under an identical design, in which pDNA is day 0 and the tested predictor is day divided by 25, so the two continuous-time fits differ only in the likelihood. A minimum total count of 30 was applied within the BARCS fit. Because all late samples are serial harvests of one culture, these fits are used as descriptive trajectory scores; their residual-t or Wald probabilities are not interpreted as independent-sample inference.

Comparator effects were not recomputed. The joint Chronos, per-endpoint MAGeCK, and per-endpoint BAGEL2 effect matrices were taken as deposited in Chronos Figshare article 14067047 [26], as were the reference essential and nonessential gene lists; the corresponding individual file identifiers are 26548532, 26548541, 30847513, 26548550, and 26548553. The latter two matrices carry one column per day; the day-25 column is used directly, so the comparator is a single observed endpoint at the deepest time point rather than a summary across days.

Two external annotations define the evaluation sets. Positives are the deposited reference essential genes. Negatives are HT-29 genes with DepMap Public 20Q2 expression below 0.5, taken from the ACH-000552 column of Figshare file 26548487, which is the expression source used for the same threshold in the deposited Chronos analysis. The copy-number audit uses the ACH-000552 column of Figshare file 26548484 [27], and correction was performed with MAGeCK 0.5.9.5's official piecewise `-cnv-norm` normalizer applied after model fitting. Because that profile was measured independently of the screen, cell-line drift is a limitation of the copy-number analysis.

The deposited workflow records the source identifier and checksum of every downloaded input, executes the official MAGeCK model and copy-number normalizer, and reproduces the tables reported here.

3.4.5 CRISPulator FACS benchmark

We used CRISPulator 0.5.1's documented interfaces for library construction, transfection, FACS selection, and sequencing [17]. The committed Julia environment pins the tagged source revision. For each simulation seed, one CRISPRn guide library was constructed and reused across four independently seeded transfection, sorting, and sequencing replicates. The default CRISPulator phenotype mixture was retained: 75% inactive, 5% negative-control, 10% phenotype-increasing, and 10% phenotype-decreasing genes. The headline screens contained 10,000 genes and five guides per gene, so 50,000 guides, which is the order of a genome-wide library rather than a pilot. The default guide-quality mixture—90% complete-knockout guides and 10% low-activity guides—was retained. Transfection representation was 500 cells per guide, Gaussian phenotype noise was 2, sorting representation was 50 cells per guide, and sequencing depth was 50 reads per guide per sample. Multiplicity of infection was 0.20 for the main analysis and 0.30 for the supplementary analysis; 0.20 is the point at which the single-integration approximation these guide-level models share is most defensible, and 0.30 stresses it. Both multiplicity of infection and the high-quality-guide fraction are exposed as simulation parameters, as is the number of genes and the number of independent screen replicates. The three genome-scale benchmark seeds were 20250724 through 20250726; the supporting 400-gene analyses below use 20250724 through 20250728.

Two BARCS gene statistics are carried through the genome-scale ablation. BARCS-ST is the historical calibrated signed-z statistic. BARCS-MOD reuses the identical guide fits after guide-dispersion moderation, so the two differ only in how the guide variance is estimated, not in the gene combiner. The exchangeable-normal, random-effects, and moderated random-effects statistics are retained for the real-data comparisons but are not carried through the genome-scale simulation, where the question is whether moderation changes the calibration–power tradeoff rather than which of several pooling rules is best. Official MAGeCK-MLE 0.5.9.5 and CRISPhieRmix are included in that run, but this restricted set is not treated as an exhaustive comparator panel. MAGeCK-MLE's ordered marker is affinely mapped onto zero–one so that the low samples of the reference replicate form the all-zero reference row the MAGeCK initializer requires. Because MAGeCK-MLE omits negative-control genes from its gene summary once their sgRNAs are declared, all methods in the genome-scale benchmark are scored on the genes every method returned a finite result for, 9,475–9,517 of 10,000 per run. The negative-control diagnostic is instead taken from each method's own output, and is therefore unavailable for MAGeCK-MLE.

We additionally performed a one-factor-at-a-time sensitivity analysis around the baseline $(\mathrm{MOI}, q, G, R) = (0.25, 0.90, 400, 4)$, where q is the high-quality-guide fraction, G is the number of genes, and R is the number of independent screen replicates. The evaluated levels were $\mathrm{MOI} \in \{0.10, 0.25, 0.40\}$, $q \in \{0.60, 0.75, 0.90\}$, $G \in \{100, 400, 1000\}$, and $R \in \{1, 3, 4, 6\}$. After removing duplicate baseline settings, this yielded ten scenarios; each used the same five prespecified seeds, for 50 simulations. One parameter changed at a time, so its main sensitivity remained interpretable. For each seed and scenario, we paired BARCS and MAGeCK-MLE results from the low–bulk–high design and reported differences as BARCS minus MAGeCK-MLE. The executable benchmark also supports the complete $3 \times 3 \times 3 \times 4$ factorial (108 scenarios and 540 simulations across five seeds) when interaction effects are the target.

The $R = 1$ setting was prespecified as a diagnostic boundary rather than a confirmatory design. Its three low–bulk–high samples and two regression coefficients leave only one residual degree of freedom. The corresponding two-tail fit has two samples and two coefficients, leaves zero residual degrees of freedom, and was therefore not fitted. All pairwise-design comparisons and the primary performance interpretation use $R \geq 3$; the one-replicate result is shown separately to expose the consequence of insufficient replication. BARCS-GC was evaluated only at $R = 1$ using Equations (30)–(33); it was not used to replace ordinary BARCS in supported replicated designs.

CRISPulator's arbitrary percentile ranges were set to $(0, 0.25)$ for low, $(0.75, 1)$ for high, and $(0, 1)$ for bulk. Thus bulk deliberately overlaps the tails and is not a third component of a multinomial partition. An input sample was generated as an independent sequencing draw from the same guide frequency vector as bulk. Low and high received phenotype scores equal to the means of their corresponding standard-normal intervals,

$$d_{\mathrm{low}} = -1.271, \qquad d_{\mathrm{high}} = 1.271,$$

while bulk received $d_{\text{bulk}} = 0$. BARCS and MAGeCK-MLE fitted this score with replicate indicators [6]. Tail-only fits removed bulk without changing the low and high counts. BARCS guide tests were scaled with the simulated negative-control guides before directional Stouffer aggregation; MAGeCK-MLE used its native gene-level Wald result. The separate 400-gene, five-seed baseline includes edgeR-QL, DESeq2, and limma–voom in addition to BARCS and MAGeCK-MLE. It is used to prevent the restricted genome-scale set from supporting a general method-ranking claim. CRISPhieRmix [28] was run on guide-level $\log_2$ fold changes from a DESeq2 Wald fit of the same design [2], which is the input its documentation specifies; DESeq2 therefore appears as a preprocessing step in the 10,000-gene run, while its native result is scored in the separate 400-gene comparison. CRISPhieRmix was run with its bimodal alternative, because this screen contains both phenotype-increasing and phenotype-decreasing genes and the one-sided default would score only one direction. Its gene effect is the mean guide $\log_2$ fold change and its local false-discovery rate stands in for a p-value wherever a ranking is required, since the model reports neither a p-value nor a signed gene effect. Reported runtimes cover the primary low–bulk–high model fit and exclude simulation, file input/output, and guide-to-gene aggregation.

Genes from the increasing and decreasing classes were positives. Ranking used the two-sided gene significance score. AUROC and average precision measure active-gene discrimination. Effect recovery is Spearman correlation between the inferred gene effect and CRISPulator's median theoretical guide phenotype among active genes. Directional recall is the fraction of active genes with gene FDR below 0.10 and an effect sign matching the simulated class. Realized FDP is the fraction of called genes that were inactive, and F1 uses active versus inactive status at the same FDR threshold.

3.4.6 simCRISPR interaction benchmark

CRISPulator assigns each guide a single effect and sorts cells on a phenotype, so it cannot generate the design BARCS is aimed at. simCRISPR [18] simulates a factorial screen instead: knockout induction crossed with treatment, propagated over several days of cell growth, with PCR amplification and sequencing applied afterwards. Every sgRNA therefore carries a knockout effect, a treatment effect, and an interaction between the two, and that interaction is a coefficient in a design matrix rather than a contrast of two arms.

Each of three seeds generated 2,000 sgRNAs, of which 300 were non-targeting and 200 were safe-harbor controls, in three independent replicates of each of the four arms, so twelve libraries and eight residual degrees of freedom. Growth was logistic. The alternative exponential model is unbounded: over five days a single guide reached a fifth of the library and library sizes spanned a 175-fold range, which does not represent a pooled screen. Initial library columns were dropped, since they are not part of the factorial contrast. Amplification used the package defaults and each library was sequenced to 500 reads per guide. BARCS fitted $\operatorname{logit}(\mu_{gi}) = \beta_0 + \beta_1 k_i + \beta_2 t_i + \beta_3 k_i t_i$ for knockout indicator k_i and treatment indicator t_i, and tested β_3.

Three denominators were compared. The library denominator is the full column total, BARCS's default. The two control denominators hold a chosen control class at a constant share, rescaled to the mean library size and rounded, so that they remain interpretable as library sizes. The two control classes are not interchangeable: safe-harbor guides are cut and carry the DNA-damage response, non-targeting guides are neither. Each control class was split in half, one half normalizing and calibrating and the other held out, so that no guide used to fit the null was also scored against it. MAGeCK-MLE received the same design matrix with an explicit interaction column, the same calibration-half controls, and one sgRNA per gene, because the simulated truth is guide-level.

CRISPhieRmix was not run on these data. It borrows strength across the guides within a gene, whereas simCRISPR assigns every sgRNA an independent interaction, so grouping guides into synthetic genes would impose the structure that the model exploits.

Every targeting guide receives a nonzero interaction, most far too small to resolve, so a magnitude was fixed before scoring: guides with $|\text{INT}| \geq 0.2$ are the positives and the held-out non-targeting guides, whose interaction is exactly zero, are the negatives. Targeting guides below that magnitude are scored for effect recovery but excluded from the detection metrics. Effect recovery is the Spearman correlation between the estimated interaction coefficient and the simulated one across all targeting guides, which needs no threshold. The held-out control call rate and the safe-harbor call rate are reported separately: the first measures false positives against a ground-truth zero, the second measures how often the cutting response alone is mistaken for an interaction.

3.4.7 Liang Cas13 longitudinal fitness benchmark

We analyzed the transcriptome-scale RfxCas13d fitness screens of Liang et al. [14] in HAP1, HEK293FT, MDA-MB-231, and THP1 cells. The library contains 56,322 guides targeting 5,496 lncRNAs, 3,156 protein-coding genes, and 1,000 non-targeting controls. Deposited Table S2 contains guide measurements at days 0, 7, and 14. The previous endpoint analysis selected only days 0 and 14; the present analysis uses all three time points.

The deposited values are fractional after median-of-ratios normalization, ComBat correction, and replicate-outlier processing. They are therefore not literal sequencing counts. BARCS requires integer observations, so each value was rounded once to the nearest pseudo-count and the identical rounded matrix was supplied to every newly fitted method. The mean absolute rounding change was approximately 0.25 and the maximum was less than 0.5 in every cell line. Consequently, this benchmark is a same-input sensitivity analysis rather than a likelihood-faithful raw-count comparison.

BARCS, MAGeCK-MLE, edgeR-QL, DESeq2, and limma–voom fitted a linear longitudinal effect with time scaled from 0 to 1 over 14 days and a replicate indicator. K562 was excluded because one deposited day-0 replicate is absent, rather than fitting an unmatched trajectory without the replicate block. BARCS guide statistics were calibrated with the non-targeting controls and combined with the prespecified directional Stouffer summary. BARCS, MAGeCK-MLE, edgeR-QL, DESeq2, and limma–voom used two-sided guide or Wald probabilities. For the four guide-level regression methods, coefficient direction was retained in the signed Stouffer score; the resulting gene probability was two-sided and was adjusted by the Benjamini–Hochberg procedure. Endpoint RRA summaries were excluded because they do not estimate the three-time-point longitudinal coefficient.

Known-essential protein-coding genes were positives. Cell-line-specific nulls were targeted lncRNAs with TPM equal to zero in both available expression assays. We reported average precision, essential-control recall at an empirical 5% null false-positive rate, the absolute deviation of the null $p < 0.05$ fraction from 0.05, and directional essential-control recall at gene FDR 0.10. A restartable raw-read pipeline is also provided for BioProject PRJNA1344834 using the stated Cutadapt anchors [29] and Bowtie `-v 1 -m 3 -best -q` [30].

3.4.8 Relationship to the legacy two-group statistic

BARCS and CB^2 share a sampling principle but not an implemented statistic. CB^2 estimates two weighted group proportions, permits separate group dispersions, forms their difference, and uses a Welch–Satterthwaite reference distribution. BARCS fits a logit coefficient with one guide dispersion across the full design and uses residual $m - q$ degrees of freedom. These choices differ in finite samples even for a binary design.

An identity-link two-cell generalized least-squares model can be constructed to reproduce the already-computed CB^2 group summaries and variances. This is an algebraic restatement of those summaries, because the working precision matrix is defined from CB^2's own weights. It is useful as a software check but is not a nesting theorem and supplies no evidence about BARCS calibration.

For small differences away from the probability boundaries, the delta method makes a logit contrast approximately proportional to a raw-proportion contrast. A stronger equivalence would additionally require the groupwise and pooled dispersion estimators to converge to a common value as the number of independent biological libraries increases. Condition-dependent dispersion violates that assumption, and deeper sequencing at a fixed number of biological units does not create the required asymptotic regime. We therefore make no formal equivalence claim. The operational generalization is limited to the mean model: BARCS can estimate continuous, adjusted, and interaction coefficients that the two-group CB^2 interface cannot express.

4 Software and usage

BARCS is distributed as an R package whose regression layer runs on base R alone, so it remains usable without a compiler. Its two inner kernels, the weighted IRLS cross-products and the symmetric solve, are implemented in RcppArmadillo [31] within the package itself and are selected automatically when the compiled library is available; the base R path is retained as an exactly equivalent fallback. The package imports CB^2 rather than vendoring it, so the original two-group functions and the FASTQ quantifier remain available and unchanged from their own package: BARCS is an additive analysis path rather than a breaking replacement.

```
library(BARCS)

# One guide: K contains guide counts; N contains sample library sizes.
fit <- bbreg(
  count   = K,
  total   = N,
  formula = ~ scale(dose) + batch,
  data    = sample_data
)
summary(fit)

# A named contrast; unspecified coefficient weights are zero.
bb_contrast(fit, c("scale(dose)" = 1))

# A complete guide-by-sample screen.
result <- bb_screen(
  counts  = count_matrix,
  totals  = N,
  data    = sample_data,
  formula = ~ scale(dose) + batch,
  term    = "scale(dose)",
  gene    = guide_annotation$gene,
  ncores  = 4
)

# Optional empirical-null scaling when prespecified negative controls exist.
# Evaluate the scale on held-out or cross-fitted controls.
result <- bb_calibrate_controls(
  result,
  control = guide_annotation$gene == "Non-Targeting Control"
)
result[order(result$fdr), ]

# Optional direct shared-effect gene inference; no Fisher/Stouffer step.
gene_result <- bb_gene_consistency(
  result,
  control = guide_annotation$gene == "Non-Targeting Control"
)
```

Column order in the count matrix must match row order in `sample_data`. The phenotype should be defined before looking at guide results, and all independent biological replicates should remain separate. For the negative-binomial benchmark, the project calls the official `mageck mle` executable with the same design matrix rather than embedding a partial MAGeCK reimplementation. On the development machine, the installed package analyzes 10,000 guides in 4.73 seconds serially and 1.36 seconds with four forked workers. The descriptive HT-29 trajectory fit processes 86,882 filtered guides across eight aggregated harvests in 21.4 seconds with four workers (4,068 guides per second); 86,840 fits reached the strict convergence tolerance, and all fits returned finite time coefficients.

Source code, executable analysis workflows, and compact benchmark outputs are available at `https://github.com/jeonglab-bcm/BARCS`. Public datasets are identified by their accession or repository record in the corresponding Methods subsections.

5 Discussion

BARCS contributes a CRISPR-oriented implementation of beta-binomial regression: immutable library totals, an arbitrary design matrix, guide-level coefficient tests, and screen-scale execution. This extends the questions available through CB^2, but it does not establish a new distribution family or a finite-sample nesting result. Beta-binomial regression is already available in `corncob` and general statistical packages, and the BARCS and CB^2 estimators differ materially.

The most important result is the calibration failure in Table 1. Under a correctly specified beta-binomial simulation, unmoderated BARCS has null type-I error 0.081 and empirical FDP 0.130 at nominal 0.05. MAGeCK-MLE is close to or below nominal in the same simulation. No guide reaches the $\widehat{\rho} = 0$ boundary, so lower-bound truncation is not the explanation in this instance. Plug-in treatment of guide-specific dispersion and the coefficient reference distribution remain plausible sources, but identifying the mechanism requires a dedicated simulation grid rather than a post hoc assertion.

The real-data diagnostics do not overturn that result. Five-fold cross-fitting in the IL2RA screen yields a held-out non-targeting rate of 0.049, showing that the control-scaling rule transfers to omitted controls at the chosen tail probability. The FACS bins are nevertheless correlated partitions of three donor pools, so a scalar correction does not make the independent-bin likelihood correct. Waterbear and MAUDE model that structure directly and recover more validated regulators. The selected 33-gene follow-up panel is also enriched by design and cannot support an unbiased genome-wide accuracy claim.

The Liang analysis demonstrates that the software can fit a three-time-point coefficient in four replicate-complete cell lines. It separates ranking from a proxy-null diagnostic and includes general RNA-sequencing comparators. BARCS has the lowest mean cell-line-specific calibration error, whereas the other methods rank essential genes more strongly. Because every method receives normalized, ComBat-corrected, rounded values, this is a robustness analysis of procedures on one transformed matrix, not evidence for either the beta-binomial or negative-binomial likelihood. The same preprocessing standard is applied to the external Chronos audit.

The HT-29 analysis has a different limitation. Its seven days are serial harvests of one passaged culture. The resulting trajectory scores are useful descriptively, but their sample-level degrees of freedom do not represent seven independent biological units. The deposited day-25 MAGeCK score has the highest AUROC and high-precision recall, so this dataset supplies no evidence that using the time course improves those endpoints.

The simulation ablations identify useful implementation choices without supporting broad competitive claims. Dispersion moderation improves BARCS within the 10,000-gene CRISPulator runs, and control-guide denominators improve interaction recovery in simCRISPR. The broader 400-gene comparison shows that edgeR-QL, DESeq2, and limma–voom can match or exceed BARCS ranking and F1, albeit at different realized FDPs. BARCS-MOD is fully defined in Equations (17)–(18), but it has not been validated on the real-data settings where calibration is problematic.

Taken together, the results support software availability and a research direction, not confirmatory deployment. A production version requires correlation-aware designs for repeated biological units and sorted bins, likelihood-compatible raw counts, dispersion inference that attains nominal error over a prespecified simulation grid, hierarchical guide-to-gene inference, and calibration evaluated only on held-out or cross-fitted controls.

6 Limitations

1. **Independent samples.** The t_{m-q} reference distribution assumes independent sequenced libraries. Repeated measures, matched samples, or multiple libraries from one biological unit require a subject-level correlation model or a blocked permutation. FACS bins from one donor are a compositional partition, not four independent biological replicates; the GSE242880 control calibration is a diagnostic sensitivity analysis, whereas a purpose-built joint model such as Waterbear represents that structure directly. Likewise, a 0–100% bulk pool overlaps its sorted tails. The CRISPulator analysis includes that requested design to measure its empirical behavior, not to redefine bulk as an independent third bin. The HT-29 days are serial harvests of one passaged culture and are therefore reported only as descriptive trajectory scores, not as t_6 inference.

2. **Enough sample-level degrees of freedom.** Read depth cannot replace biological replication. With very few samples, guide-wise $\widehat{\rho}$ is noisy and coefficient tests may be poorly calibrated. The model-based covariance treats $\widehat{\rho}$ as a fixed plug-in value, and the t_{m-q} reference does not formally propagate its estimation uncertainty. Deeper sequencing at fixed replication is not a substitute for additional independent libraries. The optional `bb_calibrate_controls()` procedure additionally requires all control guides to share the same residual degrees of freedom, as they do when every guide uses the same complete design. Guides fitted with heterogeneous designs require stratified or redesigned calibration. A control guide used to estimate a tail-quantile scale cannot also provide an unbiased evaluation of that scale; diagnostics must use held-out controls, cross-fitting, or an independent experiment. The optional `bb_gene_consistency()` statistic can recover concordant multi-guide signal when sample-level degrees of freedom are inadequate, but its empirical-null p-values measure guide consistency. They do not repair the missing biological replication and should not be described as confirmatory sample-level inference. Dispersion shrinkage across guides or permutation of the phenotype should be evaluated before treating this prototype as a production method.

3. **Mean-model form.** A single slope assumes a linear relationship on the logit scale. Residual plots should be checked. A spline or a prespecified nonlinear term is preferable when the dose response is curved. This matters directly in viability time courses: incomplete knockout can produce finite depletion saturation, which Chronos models mechanistically but a single beta-binomial slope does not.

4. **Sparse and separated guides.** Guides with all zero counts or with counts confined to one end of the phenotype range can cause boundary fits. Low-count filtering should be chosen without reference to the tested phenotype. A penalized or likelihood-ratio procedure is a useful future extension for severe separation.

5. **Compositional counts.** Guide counts share a fixed library total. The marginal beta-binomial model handles depth and guide-specific overdispersion but does not remove global composition shifts. Stable nontargeting controls and sensitivity analyses with alternative size factors are advisable. If guides are filtered for testing or benchmarking, the `totals` argument must retain the unfiltered mapped-read totals. Recomputing totals after row subsetting changes the likelihood and can materially change calibration.

6. **Guide-to-gene inference.** The BARCS regression itself is guide-level and requires no Fisher or Stouffer combination. Some replicated benchmark analyses use a directional Stouffer summary only as a transparent, explicitly labeled comparison adapter. BARCS-GC instead estimates one shared guide coefficient by inverse-variance weighting and calibrates its Wald statistic against a robust gene-level empirical null for the one-replicate diagnostic. That fixed-effect construction assumes a common gene effect, while its all-gene MAD assumes that fewer than half of genes are active. The direction-agreement output should be inspected for guide heterogeneity. We additionally evaluated BARCS-NORM, which treats the fitted guide coefficients within a gene as exchangeable draws $\widehat{\beta}_{gj} \sim N(\mu_g, \sigma_g^2)$ and tests the sample mean with a t_{m_g-1} reference because σ_g is estimated. This literal normal model was calibrated but underpowered with three to six guides: across the 50 supporting 400-gene CRISPulator runs its average precision was 0.706 and its directional

recall at FDR 0.10 was 0.159. Replacing the t reference by a plug-in normal tail would incorrectly treat the estimated σ_g as known. A production hierarchical gene test should therefore borrow variance information across genes and be benchmarked separately, ideally with held-out controls or phenotype permutations that preserve cross-guide dependence.

7. **Processed longitudinal inputs.** The Liang deposited values are normalized, batch-corrected pseudo-counts, so the fitted likelihood does not retain a literal raw-count interpretation. The three-time-point analysis is longitudinal, but raw-read confirmation remains necessary before using it as a distributional validation. Likewise, post-fit CNV correction should be accepted only after a null-gene diagnostic: it succeeds in A375 and over-corrects the longitudinal HT-29 screen.

7 Conclusion

BARCS makes library-total-conditional beta-binomial regression available for continuous, adjusted, and interaction designs in pooled CRISPR screens. That is the supported methodological contribution.

The current evidence does not establish superior calibration or generalization relative to negative-binomial methods. Unmoderated BARCS is anti-conservative in the correctly specified simulation, the strongest experimental calibration result requires cross-fitted controls, the Liang comparison uses transformed pseudo-counts, and HT-29 lacks independent longitudinal replication. Simulation ablations identify promising moderation and denominator choices, but those additions require prospective validation.

BARCS should therefore be treated as a transparent prototype for multivariable beta-binomial CRISPR analysis. Confirmatory use depends on independent biological replication, raw or likelihood-compatible counts, a correlation-aware design where needed, and an error-rate evaluation that is strictly separated from calibration fitting.

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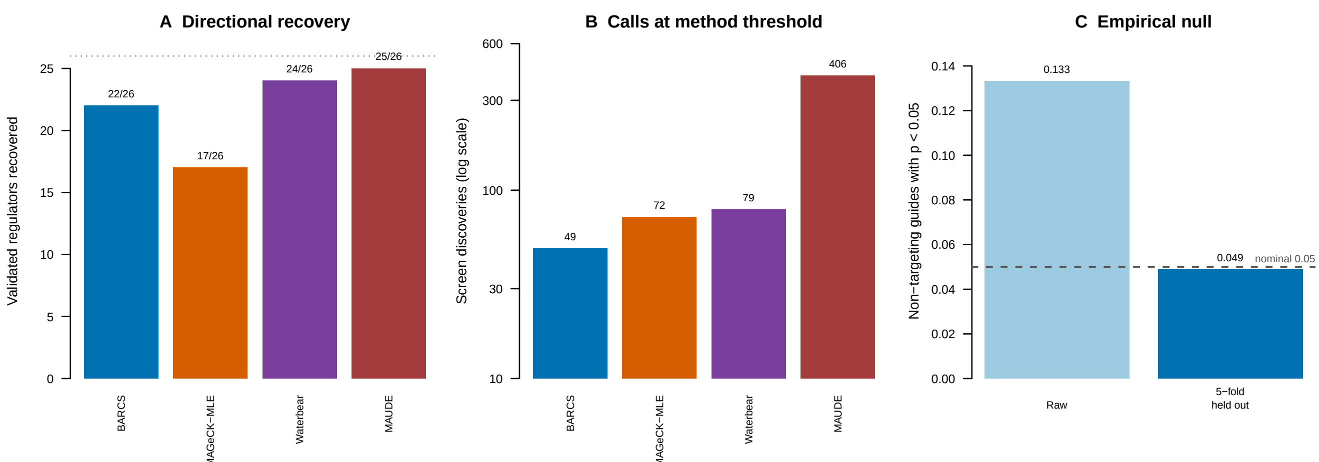
Supplementary information

Supplementary material referenced from the main text.

Supplementary Table S1: Operational definition of the problem BARCS addresses. Referenced from the Introduction.

Component	Definition
Observed data	Guide counts K_{gi}, unfiltered full-library totals N_i, and a prespecified sample design matrix X
Primary estimand	A guide-level coefficient or contrast $\mathbf{c}^\top \boldsymbol{\beta}_g$ conditional on the other columns of X
Information unit	An independent biological screen unit represented by a sequenced library; additional reads refine that unit but do not create another one
Primary output	Guide effect, standard error, t statistic, residual degrees of freedom, p-value, FDR, and dispersion diagnostics
Optional gene output	An inverse-variance shared-effect Wald statistic with empirical-null calibration; no Fisher or Stouffer step is required
Not identified	Unobserved single-cell phenotypes, dependence among bins from one biological pool, mechanistic growth dynamics, or uncertainty that would have required missing biological replicates

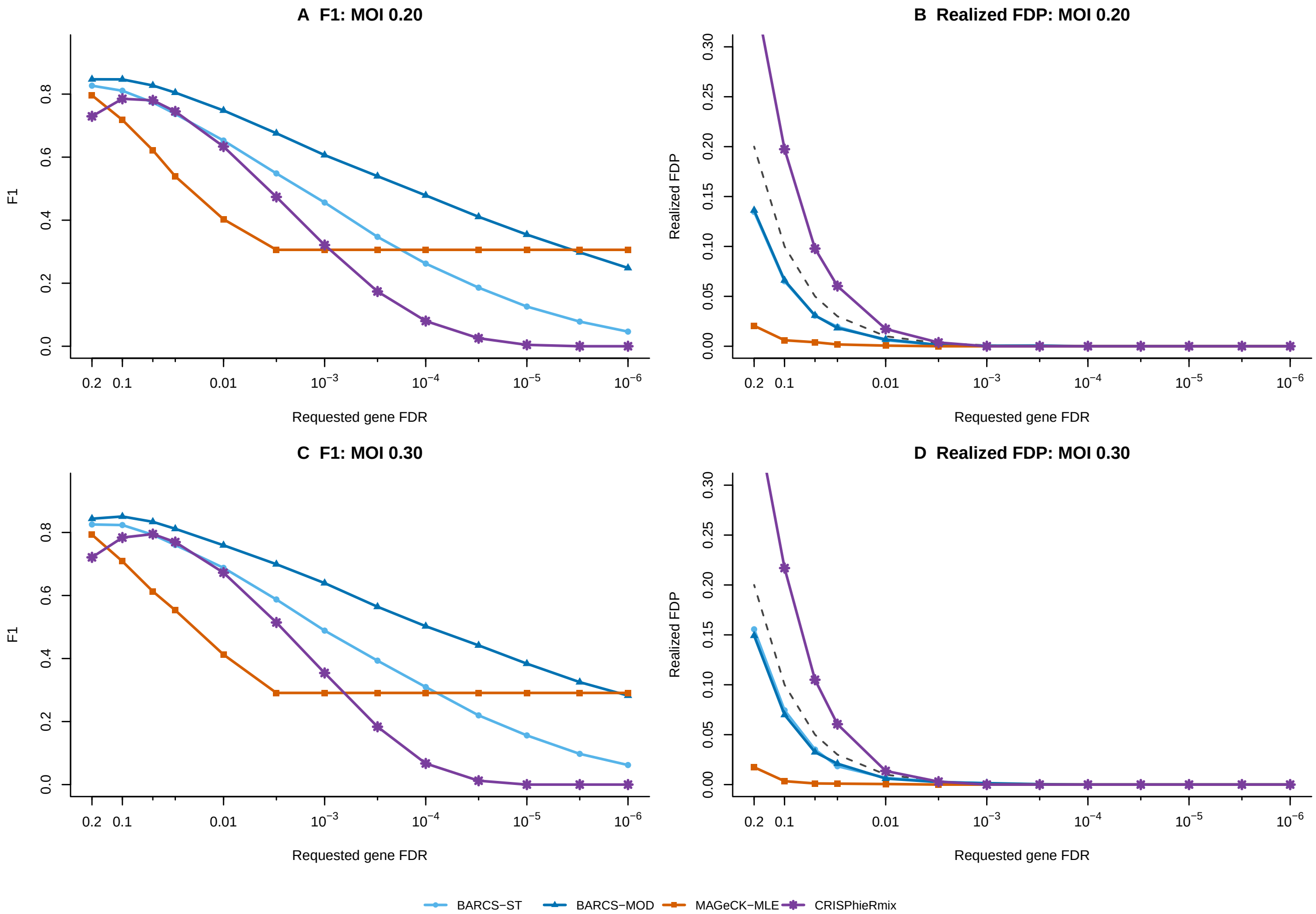


Supplementary Figure S1: Supporting ordered-bin IL2RA benchmark. (A) Directional recovery of 26 independently validated regulators. (B) Total discoveries at each method's threshold (log scale). (C) Empirical non-targeting-guide $p < 0.05$ rate before calibration and under five-fold held-out BARCS control calibration; the dashed line is the nominal rate.

7.1 HT-29 serial harvests are descriptive, not replicated longitudinal inference

The HT-29 dataset of Tzelepis et al. [25] contains one plasmid pool and guide counts from days 7, 10, 13, 16, 19, 22, and 25 of one passaged culture. The three sequencing columns at each late day were summed because they assay the same harvested population. The seven harvests are likewise repeated measurements of one biological lineage, not seven independent biological units. Referring a time coefficient to t_6 would therefore be pseudoreplication and would violate the independent-library assumption.

We retain the fitted BARCS and official MAGeCK-MLE time coefficients only as descriptive trajectory scores. They received the same eight-column matrix and numeric design. We also report the deposited all-time-point Chronos score and day-25 MAGeCK and BAGEL2 scores [12, 26, 32]. Core essential genes are positives and HT-29 genes with DepMap 20Q2 expression below 0.5 are negatives, matching the deposited evaluation.



Supplementary Figure S2: CRISPulator threshold sensitivity for the restricted genome-scale comparator set. (A,B) MOI 0.20 and (C,D) MOI 0.30. F1 is shown on the left and realized FDP on the right; the dashed diagonal denotes realized FDP equal to the requested threshold. The figure supports the BARCS-ST versus BARCS-MOD ablation. It does not include edgeR-QL, DESeq2, or limma–voom and therefore does not support a general method-ranking claim.

Supplementary Table S2: Descriptive HT-29 control-gene ranking. The BARCS and MAGeCK time rows treat serial harvests as a numeric trajectory for ranking only; their sample-level p-values are not interpreted. Recall_{90} is maximum recall at at least 90% precision, and $\text{FP}_{10\%}$ counts unexpressed genes in the lowest 10% of all evaluated effects.

Method	AUROC	PR AUC	NNMD	Recall_{90}	$\text{FP}_{10\%}$
■ BARCS time	0.9786	0.9494	−16.34	0.9037	7
■ Official MAGeCK time	0.9789	0.9534	−17.75	0.8972	5
■ Chronos joint	0.9747	0.9356	−27.88	0.8991	10
■ Deposited MAGeCK day 25	0.9792	0.9428	−14.83	0.9120	10
Deposited BAGEL2 day 25	0.9698	0.9388	−12.06	0.8630	4

The day-25 MAGeCK row has the highest AUROC and Recall_{90} in the deposited comparison. The full trajectory therefore provides no measured advantage over the endpoint on these two ranking metrics. Official continuous-time MAGeCK has the highest PR AUC, Chronos has the most negative NNMD, and BAGEL2 has the fewest unexpressed genes in the 10% tail. These differences illustrate that the metrics reward different score properties; none supplies the missing biological replication.

How a deposited time course is summarized also matters. Taking the median MAGeCK effect across all seven days instead of the day-25 column lowers AUROC from 0.9792 to 0.9742 and Recall_{90} from 0.9120 to 0.8602. Early harvests carry little depletion, so an across-day median attenuates the endpoint signal. This data-reduction effect is larger than the differences among the trajectory scores and further argues against treating this dataset as inferential evidence for one count model.

7.2 Copy-number sensitivity in the longitudinal HT-29 screen

We extracted the HT-29 profile (ACH-000552) from DepMap Public 20Q2 [27] and repeated the same copy-number audit used for A375. Before correction, effect–copy-number Spearman correlation among unexpressed genes is already small: −0.072 for BARCS, −0.075 for official continuous-time MAGeCK, −0.048 for deposited Chronos joint effects, −0.057 for the published day-25 MAGeCK effects, and +0.029 for the published day-25 BAGEL2 effects. Applying MAGeCK 0.5.9.5's official piecewise normalizer makes the association larger in absolute value rather than smaller: −0.145 for BARCS and −0.149 for MAGeCK. Thus the improvement observed in A375 does not generalize automatically.

The piecewise correction estimates one global effect–CNV trend across genes in one cell line and applies it after model fitting. When the nuisance association is weak and biological dependencies are not exchangeable across copy-number regions, the fitted trend can over-correct null genes. Chronos joint is shown without CNV correction because the published procedure requires multiple cell lines to identify common essential genes and was not applied to this single-line experiment [12]. The DepMap profile was measured separately from the original screen, so cell-line drift is an additional limitation. These results support diagnosing copy-number correction rather than applying it by default.

Additional multivariable and endpoint benchmarks. The following analyses test backward compatibility, multivariable effect concordance, copy-number sensitivity, and preprocessing contracts. They are reported separately because none is a direct test of the primary longitudinal generalization.

7.3 A multi-coefficient design on real data: GSE70038

The benchmarks so far test one coefficient at a time. GSE70038 is the case where the design matrix earns its place: 64,747 guides in 16 libraries from two glioblastoma stem-cell lines and two neural stem-cell lines, with duplicate initial and terminal samples [33]. The scientific target is four cell-line-specific terminal effects estimated *jointly* against one shared initial baseline. Four separate two-group tests cannot express that: each would re-estimate the baseline from its own pair, discarding the other twelve libraries that inform it. This is therefore not a continuous-phenotype benchmark but the multivariable one, and it is the design MAGeCKFlute documents as the canonical multi-condition layout.

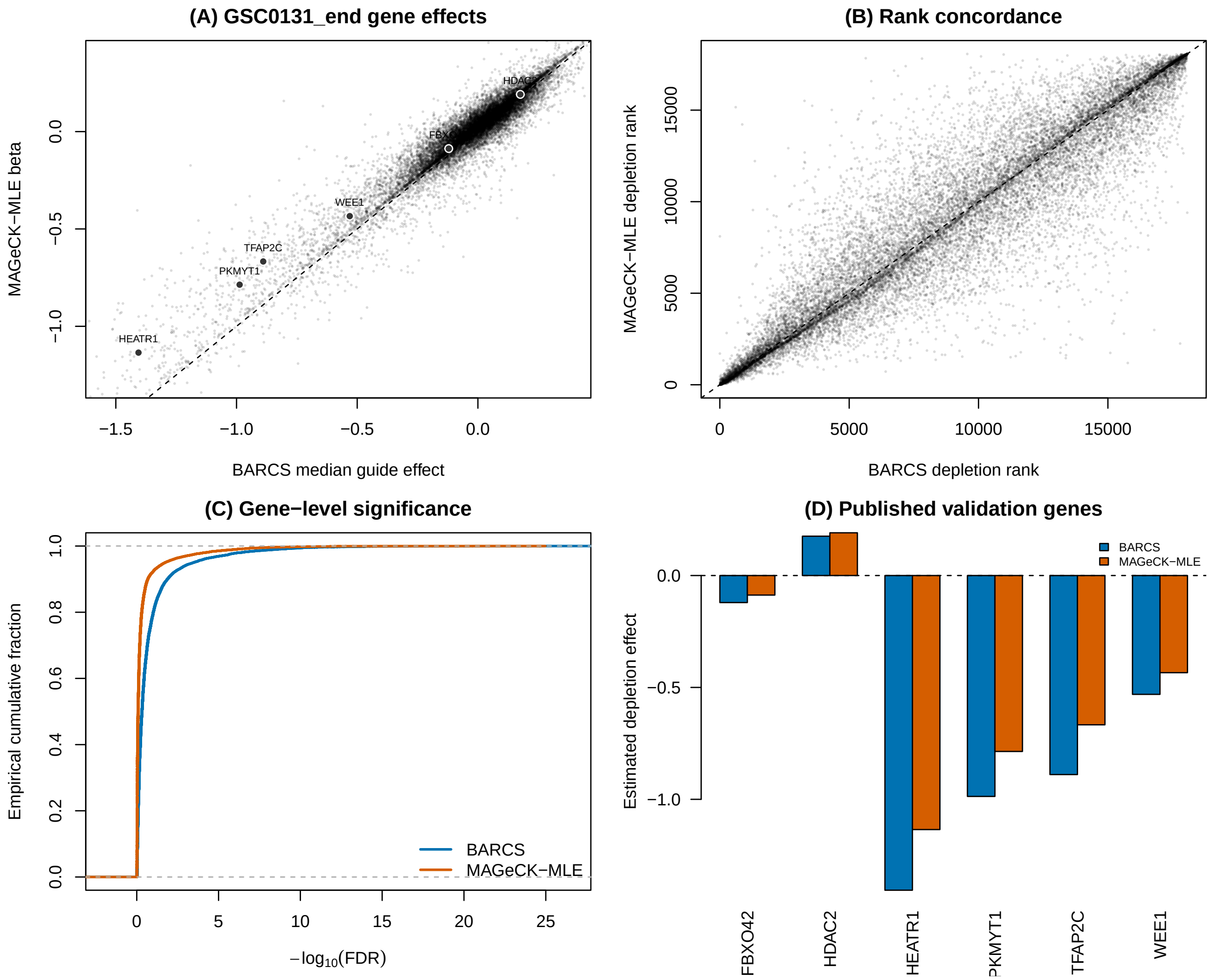
We followed the design-matrix rules in Table 5 and Box 3 of the MAGeCKFlute protocol [34]. The first column is one for every library; each initial library has zeros elsewhere; and each terminal library has one in exactly one of four cell-line columns. The same 16×5 matrix was supplied to beta-binomial regression and the official MAGeCK-MLE 0.5.9.5 executable. MAGeCK used median normalization and its Wald inference. For the beta-binomial result, guide effects were summarized by their within-gene median, and directional guide p-values were combined by Stouffer's method. This aggregation is an explicit comparison device, not a proposed replacement for MAGeCK's gene model.

The "effect rank correlation" reported below is calculated in this analysis, not copied from GEO or the MAGeCKFlute paper. Separately for each terminal coefficient, we pair all genes available from both methods. The beta-binomial effect is the within-gene median guide coefficient, and the MAGeCK effect is its official MLE beta. We assign average ranks to these two effect vectors and take their Pearson correlation, which is exactly Spearman's rank correlation. The 18,077 paired effects and their ranks for every coefficient are included with the deposited analysis output.

Supplementary Table S3: Effect concordance between ■ BARCS and ■ official MAGeCK-MLE on GSE70038. Jaccard is the overlap divided by the union of the 200 most depleted genes from each method. Higher agreement is better; column maxima are shown in bold.

Terminal coefficient	Spearman effect correlation	Top-200 Jaccard
GSC0131	**0.928**	0.550
GSC0827	0.888	0.533
NSCCB660	0.905	0.544
NSCU5	0.915	**0.562**

Concordance here is the validation rather than the finding. The comparison is not circular: beta-binomial regression conditions on each library total, whereas MAGeCK uses a normalized negative-binomial count model with guide-efficiency estimation [6]. Two models that disagree on the noise distribution, the normalization, and the guide-efficiency treatment nevertheless place the four coefficient vectors in the same order, with all four effect correlations above 0.88. That is the evidence that the model-matrix interface recovers the intended estimand on a real screen rather than a reparameterised approximation of it. These correlations measure agreement of the inferred gene ordering; they are not likelihood, calibration, or predictive-fit statistics and therefore cannot identify a better-fitting method. In GSC0131, the published convergent dependency PKMYT1 has effect −0.987 and FDR 1.87×10^{-8} under beta-binomial regression, versus beta −0.786 and Wald FDR 4.53×10^{-8} under MAGeCK-MLE. Both also recover HEATR1, TFAP2C, and WEE1 depletion. Figure S3 shows the complete GSC0131 comparison; the reproducible PDF contains one page per terminal coefficient.



Supplementary Figure S3: GSE70038 GSC0131 head-to-head. (A) Gene effects, with dark points marking the prespecified genes PKMYT1, HEATR1, TFAP2C, FBXO42, HDAC2, and WEE1. (B) Depletion-rank concordance. (C) Gene-level FDR distributions. (D) Effects for the published validation genes.

7.4 Endpoint benchmark and copy-number audit

We used the Brunello A375 negative-selection screen of Sanson et al. [35], bundled with CB^2, as a descriptive endpoint benchmark. Reference essential genes are positives and reference nonessential genes are negatives [36]. The common copy-number-complete set contains 1,506 essential and 869 nonessential genes, so its positive prevalence is 63.4%. F1 and recall in this enriched set must therefore be interpreted together with the realized false-discovery proportion.

Copy number is a gene-level property of A375 and is constant across the plasmid and terminal libraries, so it cannot be identified as a sample-level coefficient. We instead applied MAGeCK 0.5.9.5's official post-fit piecewise correction to both methods' gene effects using the A375 profile from DepMap Public 19Q3 [37, 38]. This changes effect rankings but does not refit either count likelihood.

Supplementary Table S4: Sanson A375 endpoint sensitivity analysis. Recall and gold-set FDP use nominal FDR 0.05 and a negative effect. The last column is Spearman correlation between effect and copy number among reference nonessential genes. No maximum is highlighted because the ranking confidence intervals overlap and the gold-standard set is enriched for positives.

Method	AUROC	Avg. precision	Recall	Gold-set FDP	CNV corr.
■ BARCS	0.9598	0.9786	0.7231	0.0118	−0.185
■ BARCS + CNV	0.9533	0.9762	0.7231	0.0118	−0.042
■ Official MAGeCK-MLE	0.9598	0.9795	0.7098	0.0074	−0.207
■ Official MAGeCK-MLE + CNV	0.9501	0.9762	0.7098	0.0074	−0.006

CNV correction reduces the intended nuisance association, from −0.185 to −0.042 for BARCS and from −0.207 to −0.006 for MAGeCK-MLE, but decreases AUROC for both methods. After correction, the BARCS–MAGeCK AUROC difference is 0.00316 (paired stratified-bootstrap 95% CI −0.00265 to 0.00924), and the average-precision difference is effectively zero (95% CI −0.00349 to 0.00295). At nominal FDR 0.05, BARCS has 1.33 percentage points more recall but also a larger realized FDP within the gold-standard set (0.0118 versus 0.0074). These data do not establish a ranking or calibration advantage for either count model.

7.5 External false-discovery benchmark: sensitivity to the analysis contract

A recent Chronos hit-calling preprint compared CB^2, MAGeCK, JACKS [39], DrugZ [40], and Chronos on single-condition dependency calling and differential viability [41]. The deposited notebooks and intermediate files make this comparison unusually auditable [42]. They expose two distinct observations that should not be conflated. Original CB^2 has a genuine small-sample power limitation in two-versus-two condition comparisons, but the strongest reported evidence of anti-conservative CB^2 inference is sensitive to an incompatible preprocessing step.

For guide g and sample j, CB^2 constructs the library total internally as

$$N_j = \sum_{g \in \mathcal{G}} K_{gj}.$$

The null-only Avana benchmark first restricted the count matrix to selected control genes and then ran CB^2. Consequently, its beta-binomial denominator became

$$N_j^{\text{subset}} = \sum_{g \in \mathcal{G}_0} K_{gj},$$

rather than the full sequencing depth. Filtering guides is legitimate for evaluation, but it must occur after fitting or be accompanied by immutable full-library totals. This distinction matters because within-sample rescaling leaves guide proportions unchanged while changing the beta-binomial sampling variance through N_j.

We audited this point using the deposited null-only and full-gene result tables. The published subset-matrix analysis reports 152 test-negative discoveries at FDR 0.1. Restricting the full-library CB^2 p-values to exactly the same test genes and reapplying Benjamini–Hochberg gives zero discoveries in all five cell lines (Table S5). The nominal $p < 0.05$ frequency still reaches 14.5% and 11.7% in two lines, so this correction does not establish perfect calibration. It does show that the reported 152 discoveries are not evidence about CB^2 fitted with its required full-library denominator. A second concern is that the common preprocessing applies Chronos negative-control normalization and rounding before all fits. That transformation is meaningful for Chronos, but it changes the effective sequencing depths of a method whose likelihood treats read totals as sampling information. Equal preprocessing is therefore not automatically equivalent to equal model input. The same limitation applies to the ComBat-corrected, renormalized, and rounded Liang matrix used in this manuscript. Both analyses are labeled processed-count sensitivity checks, and neither is used to validate a count likelihood.

Supplementary Table S5: Audit of the version-1 Chronos false-discovery benchmark. The Avana row uses the same test genes under two library-total contracts. The PSN1 and DeWeirdt rows are direct BARCS reruns on the deposited normalized count tables; "known" denotes the 28 literature-curated condition–gene combinations used by the preprint, not a complete truth set.

Audit target	Reported legacy result	BARCS reanalysis
Avana null-only, FDR < 0.1	152 false discoveries after pre-fit guide subsetting	0 after preserving full-library totals
PSN1 two-versus-two null split	0 gene discoveries; conservative gene p-values	4.81% guide $p < 0.05$, 4.65% after control scaling, and 0 gene discoveries
DeWeirdt differential screens	0 discoveries and 0/28 known interactions	38 discoveries and 1/28 known interactions

The differential-screen limitation is real. Original CB^2 estimates each condition separately and uses the Welch–Satterthwaite degrees of freedom

$$\nu = \frac{(\widehat{V}_A + \widehat{V}_B)^2}{\widehat{V}_A^2/(n_A - 1) + \widehat{V}_B^2/(n_B - 1)}.$$

With $n_A = n_B = 2$, $1 \leq \nu \leq 2$. Two independent variance estimates and a heavy-tailed reference distribution leave little power. In contrast, BARCS estimates one guide-level dispersion across the full design. On the PSN1 pseudo-condition split, its guide-level null is close to nominal and its negative-control scale is 1.017. This is evidence that pooled regression repairs part of the finite-sample problem.

It does not solve the full Chronos task. In an exploratory rerun of the ten DeWeirdt comparisons used for the curated-positive analysis [43], BARCS used the four terminal libraries, full count-table totals, a treatment coefficient, guide-level control scaling, Fisher gene aggregation, and BH correction. It recovered MCL1 under A-1331852 in Meljuso, but only one of 28 curated interactions overall. Thirty-seven of its 38 discoveries came from OVCAR8 under olaparib, without recovering BRCA1 or BRCA2. This pattern is consistent with a global difference in screen sensitivity rather than a failure that one additional fixed covariate can remove: with two replicates per arm, a treatment-associated quality shift is nearly collinear with treatment itself. Negative controls that are inactive in both arms also need not represent artifacts affecting strong common-essential knockouts.

The version-1 notebook raises two further reproducibility issues. First, its FDR-calibration helper accepts a requested evaluation gene set but does not use that argument; the calibration curve therefore includes training controls used by Chronos as well as held-out controls. Second, calls intended to remove MAGeCK's `common` coefficient use a non-mutating `DataFrame.drop()` without assigning the returned object. In the PSN1 calibration, this leaves 34,124 MAGeCK entries versus 17,787 entries for CB^2, JACKS, DrugZ, and Chronos. These implementation details do not invalidate the biological motivation for empirical-null hit calling, but they make the strongest method-ranking and calibration claims provisional. Similarly, treating thousands of correlated gene–screen indicators as independent observations in a t test yields uncertainty that is narrower than a screen- or cell-line-clustered bootstrap would provide.

The appropriate conclusion is therefore narrower than either "CB^2 is miscalibrated" or "BARCS solves false discovery." The regression extension repairs the mean-model and dispersion-pooling limitations, accepts CNV and other identifiable sample covariates, and gives a well-behaved null in the PSN1 check. It does not by itself supply Chronos's growth dynamics, replicate-quality terms, or empirical differential null. A production differential-screen method still needs immutable library totals, raw-count or likelihood-compatible normalization, dispersion moderation, correlation-aware gene aggregation, held-out control calibration, and blocked replicate-label permutations.